

The $L \cdot F \cdot C$ gadget pattern of $\text{MCM} \rightarrow \text{MCM-to-APN} \rightarrow \text{APN}$, categorically

Companion to `RequestProject/` — every arrow below is a `sorry-free` Lean fact

$L = \mathbb{F}_{2^n}$, $q = 2^k$, Kasami exponent $d_k = 2^{2k} - 2^k + 1$, $L_k(z) = \sum_{i < k} z^{2^i}$, $\gcd(k, n) = 1$.
H. Dobbertin, “Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets”, 1999 — Corollary 2.

1. Three gadgets, one pipeline

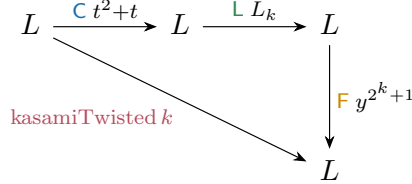
The Kasami APN proof is a composite of **three self-maps of L** .



`gadgetC, gadgetL, gadgetF : F → F` (morphisms in `Type`).

2. The chain is one categorical composite

In the category `Type` ($f \gg g = g \circ f$), the whole pipeline is a single arrow — this *is* the key identity.

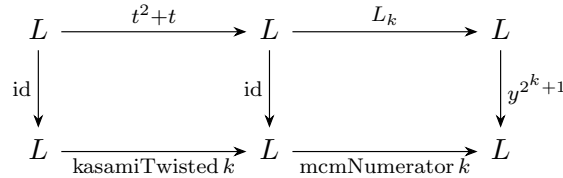


$C \gg L \gg F = \text{kasamiTwisted } k$ ($k \geq 1$). `LFCGadgets.lfc_chain, lfc_commSq`.

Key identity: $((t+1)^{d_k} + t^{d_k} + 1)(t^2+t)^{2^k} = (t^{2^k}+t)^{2^k+1}$ `MCMtoAPN.kasami_key_identity`.

3. The composite = two pasted commuting squares

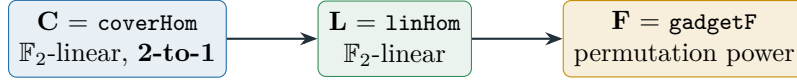
The pipeline factors as two `commSqs` glued along the MCM numerator $\text{mcmNumerator } k = F \circ L_k$.



left square = `MCMKasamiCat.mcmKasami_commSq`; right square = `mcmNumerator_commSq`; paste =
`kasamiTwisted_eq_gold_trace_artinSchreier`.

4. Which stages are $\mathbb{F}_2$ -linear (the reusable structure)

C and **L** are additive; only **F** is not. That is the whole content of the abstraction: a collapse-count on **C** rides through the additive **L** and the permutation **F**.

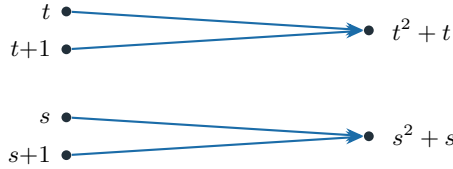


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structure LFCFactorization:  cover, lin: F →+ F;
  power, target: F → F; factors: cover ≫ lin ≫
  power = target.
Witness mcmLFC k for the MCM → Kasami chain.
  
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5. Where the APN count is born: **C** is exactly 2-to-1

The only non-injectivity in the pipeline is the covering **C**: its fibres are the pairs $\{t, t+1\}$. **L** and **F** are bijections (for valid k, n), so they transport this “2 or 0” count verbatim to the Kasami derivative — i.e. APN.

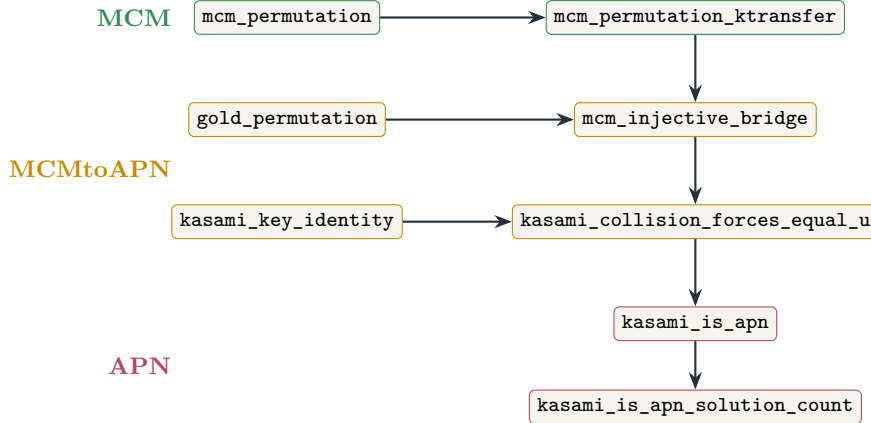


$Ct = Ct' \iff t' \in \{t, t+1\}$
covering_two_to_one

C: each hit value has exactly two preimages $\{t, t+1\}$

6. The three Lean modules, as a diagram

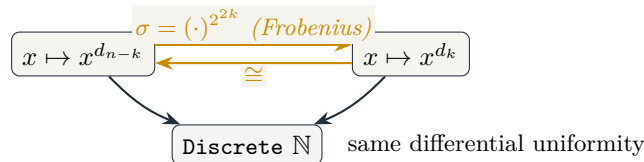
The formalisation is exactly this pipeline, split across $\text{MCM} \rightarrow \text{MCMtoAPN} \rightarrow \text{APN}$.



L·F ($\text{MCM} \circ \text{Gold}$, both bijective) = the injective bridge; **C** (Artin–Schreier, 2-to-1) = the collision collapse; together $\Rightarrow \text{kasami_is_apn}$.

7. The ambient category theory: invariance groupoid

Beyond the identity pipeline, the paper’s *even-k* reduction “replace k by $n - k$ ” is an **isomorphism in a groupoid**, and differential uniformity is a **functor** — constant on isomorphism classes.



`FrobeniusGroupoid.kasamiComplementIso`, `diffUnifFunctor`: `SelfMapGroupoid  $F \Rightarrow \text{Discrete } \mathbb{N}$` ,
`diffUnif_kasami_complement`.

gadget / pattern	map	Lean
C covering (2-to-1)	$t^2 + t$	<code>gadgetC</code> , <code>coverHom</code> , <code>covering_two_to_one</code>
L linearized trace	$L_k(z)$	<code>gadgetL</code> , <code>linHom</code>
F Gold power	y^{2^k+1}	<code>gadgetF</code> , <code>gold_permutation</code>
composite chain	$F \circ L \circ C$	<code>lfc_chain</code> , <code>mcmLFC</code>
commuting square	bridge	<code>mcmKasami_commSq</code>
APN endpoint	0 or 2 solutions	<code>kasami_is_apn(_solution_count)</code>
invariance groupoid	$k \mapsto n-k$	<code>kasamiComplementIso</code> , <code>diffUnifFunctor</code>

One line. Kasami derivative = $\underbrace{y^{2^k+1}}_F \circ \underbrace{L_k}_L \circ \underbrace{(t^2+t)}_C$: two additive stages carry the two-to-one collapse of **C** to APN.